

— A SIX-PHASE REPORT —

Predicting Pull Request Merge Outcomes.

AUTHOR / MODELING

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SOURCE

GH ARCHIVE · JULY 2015

9,269,533 PUBLIC EVENTS

BRIEF

Can public GitHub events predict whether a PR gets merged?

9.27M

PUBLIC EVENTS PARSED

195K

CLEAN PR LIFECYCLES

0.7324

ROC-AUC · BEST MODEL

0.99

NON-MERGE PRECISION @ TOP 1%

A leakage-aware ranking model trained on PR-opening signals and pre-opening activity. Designed as a *triage tool*, not an automated decision system.

PHASES OF WORK

A linear story, told in six acts.

PHASE 01 Business Understanding Why merge prediction matters for triage.	PHASE 02 Data Understanding GH Archive event stream, 14 event types.	PHASE 03 Data Preparation Leakage-aware PR-level table, 26 features.	PHASE 04 Modeling Four families. Linear, tree, boosted, dummy.	PHASE 05 Evaluation Ranking, risk@k, threshold analysis.	PHASE 06 Deployment Triage support — not auto-decision.
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TOOLING

Python · Polars · pyarrow · scikit-learn · matplotlib.

SAMPLE WINDOW

Hourly GH Archive files, continuous July 2015.

COST ENVELOPE

≈ \$688 . 40 · 130 hrs of analyst time.

01

Business Understanding.

MAINTAINERS DROWN IN PULL REQUESTS. CONTRIBUTORS GUESS AT THE RULES. WE ASK: DO EARLY, PUBLIC SIGNALS ALREADY SAY WHICH PRS WILL LAND?

1.1 OBJECTIVES

Rank newly opened PRs by merge likelihood — before maintainers spend a minute on them.

The model is a *prioritization layer* on top of GitHub's firehose. It does not reproduce maintainer reasoning; it surfaces probable outcomes from public signals.

DECISION-SUPPORT VALUE

For a team reviewing 500 PRs/month, even a 10% triage saving recovers $\approx \$1,512$ / yr at \$30/hr.

1.3 DATA-MINING GOALS

- G · 01 Convert raw GH Archive events into a PR-level supervised dataset.
- G · 02 Predict $P(\text{merged} = 1 \mid X)$ as binary classification.
- G · 03 Compare linear and non-linear models under one shared split.
- G · 04 Identify which pre-opening signals drive merge ranking.
- G · 05 Keep all features compliant with PR-publication-time information.

02

Data Understanding.

A FLAT ANALYTICAL EVENT TABLE – JSON PAYLOADS
PRESERVED, TOP-LEVEL FIELDS LIFTED, 14 EVENT TYPES
CATALOGUED, LIFECYCLE COVERAGE VERIFIED.

2.2 EVENT MIX

Top event types in the converted sample.

EVENT TYPE	COUNT	SHARE
PushEvent	4,417,737	47.7%
CreateEvent	1,329,287	14.3%
IssueCommentEvent	886,068	9.6%
WatchEvent	826,509	8.9%
PullRequestEvent	470,260	5.1%
Other event types	1,339,672	14.4%

2.4 QUALITY CHECKS

What the audit confirmed.

OK	Event volume valid — 9,269,533 records.
OK	Merge label present in all closed PR payloads.
LIM	Lifecycle coverage restricted — 195,425 usable PRs.
DROP	Author-association field constant; excluded from features.
CTRL	Closure-time fields kept only for target / diagnostics.

Action distribution within PullRequestEvent: 238,541 opened · 228,072 closed · 3,647 reopened.

03

Data Preparation.

ONE ROW PER PULL REQUEST. 26 NUMERIC FEATURES. STRICT
LEAKAGE POLICY: NO CLOSURE-TIME FIELDS, NO SAME-DAY
FUTURE EVENTS.

Five steps. One leakage-aware table.

STEP 01 Select Retain PRs with both opened & closed events containing merge label.	STEP 02 Clean Sort by timestamp, keep first opening / first closing per PR id.	STEP 03 Construct Target merged $\in \{0,1\}$; merge rate 0.8574.	STEP 04 Integrate Join opening + closure + previous-day repo / actor history.	STEP 05 Format 26 numeric features × 195,425 rows. Median-imputed.
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OPENING-TIME FEATURES

commits · additions · deletions · changed files · opening hour · weekday · weekend · draft · log-size variants.

REPOSITORY HISTORY

Previous-day repo events, PRs, issue-comments, push events, cumulative actor-days.

ACTOR HISTORY

Previous-day actor activity, PR volume, comment volume, push activity, repo-days.

LEAKAGE RULE

For a PR opened on date d, every event from date d is excluded from history features. Closure-time fields are never inputs.

04

Modeling.

DUMMY. LOGISTIC REGRESSION. RANDOM FOREST. HISTOGRAM
GRADIENT BOOSTING. SAME FEATURES. SAME SPLIT.
STRATIFIED 75 / 25.

4.4 ASSESS

HistGradientBoosting wins ranking.

F1 alone is misleading under 0.857 base rate — ROC-AUC and PR-AUC tell the real story.

Hist Gradient Boosting		0.7324
Random Forest		0.7271
Logistic Regression		0.6547
Dummy (most frequent)		0.5000

Bars scaled by ROC-AUC.

4.4 TOP PERMUTATION IMPORTANCE

Context features beat PR-size features.

FEATURE	MEAN Δ ROC-AUC
repo_push_events_before_day	0.0644
repo_actor_days_before_day	0.0579
actor_push_events_before_day	0.0368
repo_issue_comment_share_before_day	0.0330
actor_pr_events_before_day	0.0171
repo_pr_share_before_day	0.0162
commits	0.0149

Repository workflow + contributor context > raw PR size.

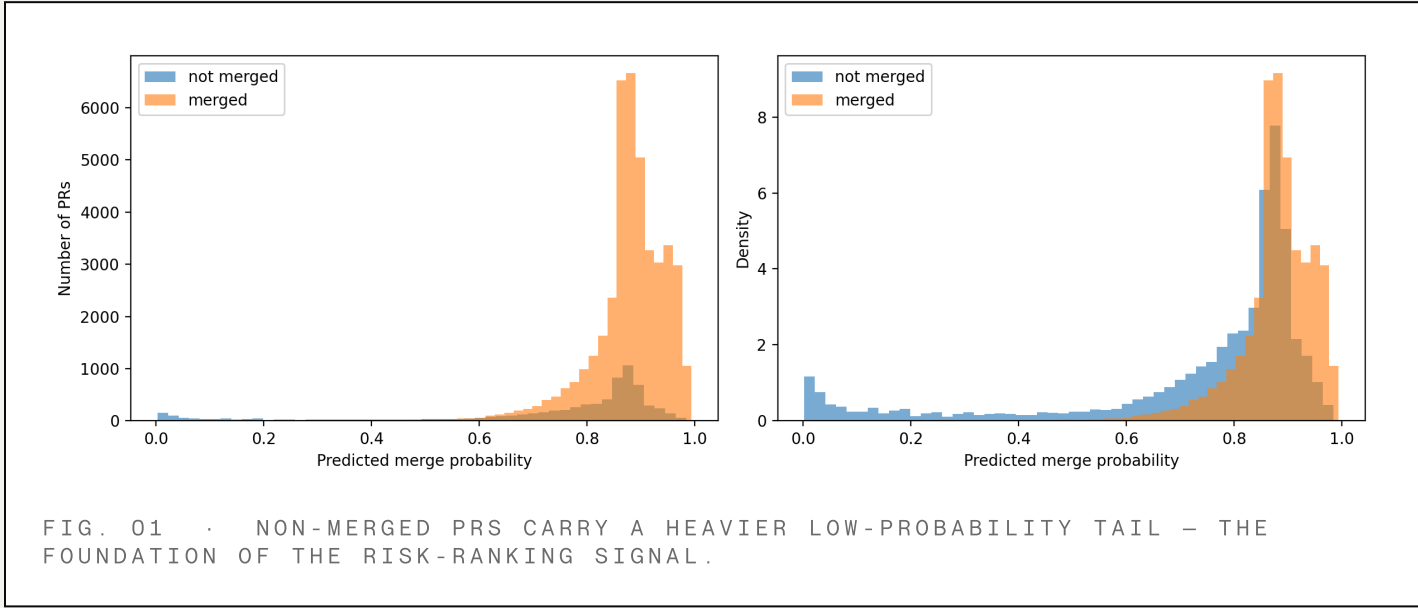
05

Evaluation.

THE MODEL IS NOT A HARD CLASSIFIER AT THRESHOLD 0.5. IT IS A RISK-RANKING INSTRUMENT – AND AT THE TOP 1% IT IS ALMOST SURGICAL.

5.1 RESULT SHAPE

Predicted merge-probability distribution.



5.1 RISK PRECISION @ K

The top-ranked tail is mostly non-merged.

RISK-RANKED SUBSET	NON-MERGE PRECISION	BASELINE RATE
Top 1%	0.9939	0.1426
Top 5%	0.6257	0.1426
Top 10%	0.4581	0.1426
Top 20%	0.3325	0.1426

In the top **1%** highest-risk PRs, **99.4%** are actually non-merged — **7×** the baseline rate.

Risk = $1 - P(\text{merged})$. Threshold analysis confirms the precision/recall trade: at risk 0.20 the model flags 15% of PRs at non-merge precision 0.38.

06

Deployment.

NO LIVE INTEGRATION. THE DELIVERABLE IS A REPRODUCIBLE ANALYTICAL ARTIFACT: REPORT, CODE, FIGURES – AND A CLEAR SCOPE FOR WHAT THE SCORE MAY AND MAY NOT DO.

6.1 RECOMMENDED USE

A maintainer dashboard, not a merge gate.

SHOW	Predicted merge probability + non-merge risk score.
SHOW	Most important feature groups: repo activity, contributor history.
WARN	Data limitations and model scope, on-screen.
NEVER	Auto-reject, hide, or deprioritize a contributor.

6.4 LESSONS & NEXT STEPS

What we proved — and what's next.

PROVED

Public, pre-opening event signals carry useful ranking signal for PR merge outcomes. Repository & contributor context dominate raw PR size.

OPEN

Longer archive window · exact-timestamp history · early-interaction features (first 6–24h) · time-based validation · richer GitHub fields.

FINAL POSITION

A proof-of-concept ranking model for PR outcome risk — useful for prioritization, insufficient for automation.

— THANK YOU

Questions welcome.

Public-event mining offers a leakage-aware
path to PR triage signal — provided we ship it
as a tool, not a verdict.